

# Jesús M. Rueda-Becerril | Ph.D.

Seattle, WA | jm.ruebe@gmail.com | +1 (765) 430-2330 | linkedin.com/in/jeruebe | altjerue.github.io  
github.com/altjerue | orcid.org/0000-0003-1988-1912 | Google Scholar

## Professional Summary

---

Computational Scientist with a PhD in Physics and 10+ years developing high-performance numerical algorithms and physics-based simulation software. Expertise spanning multiscale/multiphysics modeling, scientific computing, and scientific machine learning. Proven track record publishing peer-reviewed research, obtaining competitive grants, and collaborating across interdisciplinary teams to solve complex computational problems in physics.

## Skills

---

**Languages:** Python, C/C++, Fortran 95, R, SQL, Shell, Rust, Java, Julia

**ML & Data:** Scientific ML, PyTorch, TensorFlow, scikit-learn, NumPy, SciPy, Pandas, Astropy, ML-assisted parameter estimation, gradient-based optimization, SVM, random forest, neural networks

**Geospatial & Remote Sensing:** Sentinel-1/2 multispectral processing, raster/vector data pipelines, QGIS, geospatial feature engineering for ML

**HPC & Parallel:** OpenMP, MPI, OpenACC, SLURM, PBS, HDF5, large-scale parallel CPU computing

**Simulation:** Physics simulation (multiscale/multiphysics modeling, radiative transfer, relativistic MHD, Lagrangian particle dynamics), numerical PDEs (advection-diffusion, Fokker-Planck), finite difference/volume methods, numerical linear algebra, synthetic dataset generation, uncertainty quantification, parameter estimation

**Tools:** Git, Docker, Azure, Jenkins, Kafka, Jira, QGIS, Paraview, VisIt

## Education

---

### Ph.D. in Physics

*Universitat de València, Spain*

ADVISOR: Prof. Miguel A. Aloy & Dr. Petar Mimica.

THESIS: *Numerical treatment of radiation processes in the internal shocks of magnetized relativistic outflows*

**Oct. 2011 – Jul. 2017**

Excellent *cum laude*.

### M.Sc. in Physics

*Universidad Michoacana de San Nicolás de Hidalgo, Mexico*

ADVISOR: Prof. José A. Cervera

THESIS: *Study of TOV stars with the SPH method*

**Aug. 2009 – Sep. 2011**

### B.Sc. in Physics

*Universidad Autónoma del Estado de México, Mexico*

ADVISOR: Prof. Francisco S. Guzmán

THESIS: *Numerical solution of null geodesics for the generation of gravitational lenses produced by spherically-symmetric and static spacetimes*

**Aug. 2004 – Dec. 2008**

“Dr. Juan Josafat Pichardo” Award

## Experience

---

### Independent Contractor

*Mercor, Seattle, WA, USA*

○ Provide expert-level knowledge in Physics, Astrophysics, and Mathematics to improve models for a top AI research lab

○ Assess model reasoning quality, identify failure modes, and provide structured feedback to improve scientific accuracy

○ Apply understanding of mathematical foundations to benchmark model performance, and quality assurance on complex STEM problems

**Jan 2026 — Present**

### Spatial Data Scientist

*TealWaters, Seattle, WA, USA*

○ Lead project to optimize code for topographic terrain modeling and elevation derivative calculation.

○ Engineered Python tools for geospatial data processing and analysis, utilizing QGIS for visualization and analysis.

○ Utilize Wetland Intrinsic Potential (WIP) tool with random forest for wetland probability mapping in Skykomish watershed.

○ Performed EDA, feature engineering, and geospatial data preparation, including multispectral (Sentinel 1/2) processing for ML land-cover and wetland probability models.

○ Share knowledge and results with managers and decision-makers, train team members on WIP tool usage, and collaborate with software engineers and scientific team to transition prototypes into production.

**May 2025 — Jan 2026**

### Software Engineer

*Paychex, Rochester NY, USA; based in Seattle, WA, USA*

**Apr 2022 – Jan 2024**

- Developed Java Kafka consumers for high-throughput data transfer across enterprise databases, and deployed to production via Jenkins and OpenShift using CI/CD pipelines.
- Developed Python tests and CI pipelines ensuring software quality standards.
- Conducted SQL data preparation, validation, and analysis from Oracle EBS datasets.

#### Postdoctoral Research Associate

Feb 2021 – Apr 2022

*Rochester Institute of Technology, Rochester, NY, USA*

- Led a team of specialists on a NSF-sponsored project to upgrade the C code **PatchworkMHD** to perform HPC simulations using state-of-the-art numerical techniques to model supermassive black hole binaries.
- Implemented a new feature (black hole spin) to **PatchworkMHD**, making more realistic binary black hole simulations without impacting runtime.
- Designed the experiments and evaluated state-of-the-art mathematic and numerical algorithms implemented in **PatchworkMHD** by running simulations at *Frontera* supercomputer (TACC, UT at Austin).
- Worked in a detail-oriented manner to successfully benchmark and identify performance optimization opportunities of the scientific code.
- Supervised a graduate student applying gradient descent optimization to fit Paramo model parameters to blazar spectral energy distributions observed by the Fermi-LAT telescope, demonstrating ML-assisted parameter estimation workflows for complex astrophysical systems.
- Participated in a multi-institutional collaboration to study binary Neutron Star mergers through HPC simulations, resulting in 2 publications that provided critical breakthrough insights of the physics underlying these events.
- Published 3 co-authored papers and mentored graduate students (2 Ph.D.).

#### Postdoctoral Research Fellow

Oct 2018 – Nov 2020

*Purdue University, West Lafayette, IN, USA*

- Developed the open-source code, **Paramo**, a numerical code in Fortran 95 optimized with OpenMP to perform radiative transfer simulations in relativistic astrophysics scenarios.
- Served as a Co-I on two NASA Fermi Guest Investigator Program proposals, one granted. The awarded grant helped with research on our understanding of the two main types of blazars, identifying that important physical constraints applied to both objects.
- Developed Python tools to calculate the loss of energy due of high-energy particles due to interactions spectrum and evolution in the context of gamma-ray burst afterglows by developing sophisticated numerical integration, and OpenMP optimized features to **Paramo**.
- Collaborated with a group of multidisciplinary scientists to develop Python scripts for statistically modeling the COVID-19 outbreak in Mexico and helped create scientific infographics and blogposts for Spanish-speaking populations to reduce the spread of misinformation.
- Published 1 first-author and 1 co-authored paper and mentored three graduate students (1 M.S. and 2 Ph.D.).

#### Postdoctoral Research Fellow

Jan – Sep 2018

*Universidad Michoacana de San Nicolás de Hidalgo, Morelia, Michoacan, Mexico*

- Contributed to a Python pipeline to process 2D simulation images from the GRTrans numerical code, converting outputs to grayscale, normalizing pixel intensities, and extracting spatial features to generate training data for an SVM classifier predicting black hole spin from radio image morphology.
- Developed an open-source data analysis and visualization tool in Python to provide any user with accessible tools to calculate radiative transfer phenomena (spectra and light-curves) in relativistic astrophysics.
- Organized a workshop to train graduate students in the use of the high-volume data storage tool HDF5.

#### Graduate Research Assistant

Oct 2011 – Jul 2017

*Universitat de València, Burjassot, Valencia, Spain*

- Independently developed Shell and Python scripts to build pipelines to run simulation of the radiative transfer code **C-SPEV** and perform data processing of output datasets in HDF5 format, ensuring data quality and integrity for downstream analysis and model fit with observations.
- Developed Python scripts to perform exploratory data analysis on multi-wavelength observational datasets from the NASA Fermi-LAT telescope and the Very Large Baseline Array (VLBA), including sky-coordinate-based source cross-matching across catalogs and log-parabola spectral fitting to characterize blazar emission.
- Independently constructed models from **C-SPEV** simulations for curve fitting, pattern recognition, and prediction of data from NASA telescopes.
- Conducted multiple analyses to identify patterns in spectra and light-curves that allowed the quantification of magnetization of plasma in blazars.
- Implemented sophisticated numerical tools and data handling to **C-SPEV** that could calculate both discrete and continuous spectra from particle distributions with arbitrary shape, without impacting simulation runtime.
- Published 2 first-author papers.

## Publications

### Articles

- [12] Winkowski, S., **Rueda-Becerril, J. M.**, & Willing, C., *Using Numerical Simulations of Wind on Hilly Terrain to Module Propagule Dispersal*, in prep.

- [11] Holzman, L., **Rueda-Becerril, J. M.**, Creamer, C., & Willing, C., *Random Forest Link Landscape Position to Drought Resilience in Giant Sequoia Forests*, in prep.
- [10] **Rueda-Becerril, J. M.**, & Winkowski, S., & Willing, C. *WindsOfChange: A New Toolkit for Modeling Spore Dispersal in Hilly Terrains*, in prep.
- [9] Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Tleco: A Toolkit for Modeling Radiative Signatures from Relativistic Outflows*, ApJ **976**, 182, (2024), arXiv:2405.17581.
- [8] Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Balancing Turbulent Heating with Radiative Cooling in Blazars*, MNRAS **513**, 5766–5779, (2022), arXiv:2201.07790.
- [7] Lopez-Armengol, F. G., Etienne, Z. B., [...], **Rueda-Becerril, J. M.**, [...] *Handing off the outcome of binary neutron star mergers for accurate and long-term postmerger simulations*, Phys. Rev. D **106**, 083015, (2022), arXiv:2112.09817
- [6] Murguia-Berthier, A., Noble, S., [...], **Rueda-Becerril, J. M.**, [...] *HARM3D+NUC: A New Method for Simulating the Post-merger Phase of Binary Neutron Star Mergers with GRMHD, Tabulated EOS, and Neutrino Leakage*, ApJ **919**, 95, (2021), arXiv:2106.05356
- [5] **Rueda-Becerril, J. M.**, Harrison, A. O. & Giannios, D. *Blazar jets launched with similar energy per baryon, independently of their power*, MNRAS **501**, 4092–4102, (2021), arXiv:2009.02273.
- [4] Zhang, H., Christie, I., Petropoulou, M., **Rueda-Becerril, J. M.** & Giannios, D. *Inverse Compton Signatures of Gamma-Ray Burst Afterglows*, MNRAS **496**, 974–986, (2020), arXiv:1910.14049.
- [3] **Rueda-Becerril, J. M.**, Mimica, P. & Aloy, M. A. *On the influence of a hybrid thermal–non-thermal distribution in the internal shocks model for blazars*, MNRAS **468**, 1169–1182, (2017), arXiv:1612.06383.
- [2] **Rueda-Becerril, J. M.**, Mimica, P. & Aloy, M. A. *The influence of the magnetic field on the spectral properties of blazars*, MNRAS **438**, 1856–1869 (2014), arXiv:1310.5441.
- [1] Guzmán, F. S. & **Rueda-Becerril, J. M.** *Spherical boson stars as black hole mimickers*, Phys. Rev. D **80**, 084023 (2009), arXiv:1009.1250.

## Proceedings.....

- 5. **Rueda-Becerril, J. M.** *A numerical approach for radiative cooling in relativistic outflows*, Astron. Nachr., 9th International Workshop on Astronomy and Relativistic Astrophysics: from Quarks to Cosmos **342**, 277–282, (2021), arXiv:2011.13797.
- 4. **Rueda-Becerril, J. M.**, Harrison, A. O. & Giannios, D. *The blazar sequence revised*, Astron. Nachr., 9th International Workshop on Astronomy and Relativistic Astrophysics: from Quarks to Cosmos **342**, 147–152, (2021), arXiv:2011.13805.
- 3. **Rueda-Becerril, J. M.**, Mimica, P. & Aloy, M. A. *Numerical simulations of the internal shock model in magnetized relativistic jets of blazars*, PoS(SWIFT 10) **233**, 159 (2014), arXiv:1502.07882.
- 2. **Rueda-Becerril, J. M.**, Mimica, P., Aloy, M. A. & Aloy, C. *Numerical study of broadband spectra caused by internal shocks in magnetized relativistic jets of blazars*, EPJ Web Conf. **61**, 02007 (2013), arXiv:1309.4612.
- 1. Mimica, P., Aloy, M. A., **Rueda-Becerril, J. M.**, Tabik, S. & Aloy, C. *Numerical simulations of dynamics and emission from relativistic astrophysical jets*, J. Phys.: Conf. Ser **42**, 012001 (2013), arXiv:1211.1794.

## Research Grants

### NASA Fermi Cycle-12 Guest Investigator Program

A simple model to understand the blazar sequence, PI: Giannios, D., Co-I: **Rueda-Becerril, J. M.**

Grant #121077

2019

## Grants and Fellowships

**Oct. 2018 – Nov. 2020: Fellowship** from the Mexican Federal Government for international postdoctoral studies awarded by the National Council of Science and Technology (CONACyT).

**Jan. – Sep. 2018: Fellowship** from the Mexican Federal Government under the *Program for the Professional Development of Higher Education Institutions*, awarded by the Secretariat of Public Education.

**Sep. 2014 – Aug. 2016: Fellowship** from the Mexican Federal Government to study abroad awarded by the National Council of Science and Technology (CONACyT).

**Oct. 2011 – Jun. 2014: Fellowship** *Santiago Grisolia* awarded by the Council of Education, Research, Culture and Sport of the Valencian Community, Spain.

**Sep. 2009 – Aug. 2011: Fellowship** for MSc studies at the Institute of Physics and Mathematics, Universidad Michoacana de San Nicolás de Hidalgo, granted by the Mexican Council of Science and Technology (CONACyT).

**Jun. – Aug. 2007: Fellowship** for a temporary stay (3 months) in a national research center under the *XVII summer of scientific investigation program* awarded by the Mexican Academia of Science.

## Projects

---

### Scientific ML (Python, scikit-learn, PyTorch)

Jan 2026 — Present

*Blazar Population ML*

*GitHub*

- Applying classical ML and Bayesian inference to the Fermi-LAT 4LAC-DR3 catalog (3,407 sources) to recover physical jet parameters from multi-wavelength observational data.
- Building a neural network to learn the mapping from physical jet parameters to observable SEDs accurately enough to compare with Tleco results.
- Developing forecasting pipeline for blazar gamma-ray flares using a time series foundation model, fine-tuned with synthetic light curves generated with Tleco.

### Open-Source Lagrangian Particle Simulation Library (R/C++)

Aug 2024 — Present

*WindsOfChange*

*GitHub*

- Developing C++ library implementing Lagrangian particle dispersion, atmospheric turbulence, and fluid flow over complex hilly terrains with forest canopy, based on analytical models (Massman & Weil 1999; Finnigan & Belcher 2004).
- Parallelized particle tracking using RcppParallel and OpenMP; designed for scalability to tens of thousands of particles across kilometer-scale domains.

### Open-Source Physics Simulation Library (Python/Rust)

Jan 2024 — Sep 2024

*Tleco*

*GitHub*

- Multiscale/multiphysics simulation toolkit combining Rust and Python interfaces for modeling relativistic particle distributions via Fokker-Planck equations coupled with radiative transfer.
- Implements numerical PDE solvers, particle distribution algorithms, and spectral calculation methods for relativistic plasma physics.

### Open-Source Physics Simulation Code (CONAHCYT Funded)

Oct 2018 — Apr 2022

*Paramo*

*GitHub*

- Modular physics simulation library (Fortran 95 + OpenMP) modeling relativistic particle radiation; predecessor to Tleco.

### Large-Scale MHD Simulation Code (NSF Project)

Feb 2021 — Apr 2022

*PatchworkMHD*

- Led upgrade of production C-based MHD simulation code for HPC environments; added physics features while maintaining performance at Frontera supercomputer scale.

## Awards

---

**Marcos Moshinsky Award:** for *Best Poster* presented at the IWARA 2020 Video Conference, Mexico City, 6 – 12 September 2020.

**Lic. Juan Josafat Pichardo Cruz Award:** for finishing and defending a licentiate thesis within a year after completing the undergraduate credits, granted by the Universidad Autónoma del Estado de México, 2009.

## Invited Talks

---

### The Role of Machine Learning in Environmental Science

*Seminar on Climate Change Ecology, University of Washington, Seattle, WA, USA, March, 2024*

### Morphology of the spectra from numerical simulations of the internal shocks model for blazars

*Astrophysics Seminar, Purdue University, West Lafayette, IN, USA, February 4, 2019*

### Numerical simulations of the internal shocks model in magnetized relativistic jets of blazars

*DATA group weakly Seminar, Instituto de Astronomía, UNAM, Mexico City, Mexico, June 19, 2018*

### Numerical treatment of non-thermal radiation in the internal shocks model for blazars

*Weekly Seminar, Instituto de Física y Matemáticas, Morelia, Mexico, March 2, 2018*

### Numerical simulations of the internal shock model in magnetized relativistic jets of blazars

*IVICFA's Fridays: Computation in Physics, IFIC, Paterna, Spain, October 17, 2014*

## Meetings and conferences

---

### Contributed Talks

#### Simulations of supermassive binary black holes accretion dynamics in the spinning case

*APS April Meeting 2022, New York City, NY, April 9–12, 2022*

## **A numerical approach to the Klein-Nishina corrections of radiative cooling in relativistic outflows**

*APS April Meeting 2021, April 17–20, 2021*

## **The blazar sequence revised**

*9th International Workshop on Astronomy and Relativistic Astrophysics, Video Conference, September 6–12, 2020*

<https://www.youtube.com/watch?v=BAZNWLNT69M>

## **Influence of the magnetic field on the spectral properties of blazars in the internal shocks scenario**

*Extreme-Astrophysics in an Ever-Changing Universe, Ierápetra, Greece, June 16–20, 2014*

## **Numerical study of broadband spectra caused by internal shocks in magnetized relativistic jets**

*XXXIV Biennial meeting of the Royal Spanish Society of Physics, Valencia, Spain, July 15–19, 2013*

## **Poster Sessions**.....

### **A numerical approach for radiative cooling in relativistic outflows**

*9th International Workshop on Astronomy and Relativistic Astrophysics, Video Conference, September 6–12, 2020*

Marcos Moshinsky Award for Best Poster. <https://www.youtube.com/watch?v=0TJiKg7k0PI>

### **Numerical simulations of the internal shock model in magnetized relativistic jets of blazars**

*Swift: 10 years of Discovery, Rome, Italy, December 2–5, 2014*

### **Numerical study of broadband spectra caused by internal shocks in magnetized relativistic jets**

*The Innermost Regions of Relativistic Jets and Their Magnetic Fields, Granada, Spain, June 10–14, 2013*

## **Teaching & Mentoring Experience**

---

**Zachary Davis** [8, 9]

*Graduate student, Department of Physics and Astronomy, Purdue University*

**Mentoring**

*2018 – 2022*

**Amanda O. Harrison** [5]

*Graduate student, Department of Physics and Astronomy, Purdue University*

**Mentoring**

*2018 – 2020*

**Hao Zhang** [4]

*Graduate student, Department of Physics and Astronomy, Purdue University*

**Mentoring**

*2018 – 2019*

**Thermodynamics (Graduate Level)**

*Instituto de Física y Matemáticas, Universidad Michoacana de San Nicolás de Hidalgo*

**Substitute Instructor**

*Jun 2018*

## **Professional Development**

---

### **High Performance Computing on Frontera**

*Jason Allison et al., TACC, Austin, TX, USA, May 20, 27 and June 3, 2021*

*Lecture*

### **Writing Winning Grants**

*Dr. Lauren Broyles, Purdue University, West Lafayette, IN, USA, November 7, 2019*

*Lecture*

### **XSEDE HPC Workshop: Summer Boot Camp**

*John Urbanic, Purdue University, West Lafayette, IN, USA, June 3 – 6, 2019*

*Workshop*

### **Data Analysis and Machine Learning with Python**

*Dr. Alejandro Torres, Universitat de València, Burjassot, Spain, February 7 – 16, 2017*

*Workshop*

### **Numerical Relativity Simulations of BBH Coalescence Using the Einstein Toolkit**

*Dr. Vassilios Mewes, Universitat de València, Burjassot, Spain, July 6 – 7, 2016*

*Workshop*

No. of hours: 8

### **The Universe in the Light of PLANCK and BICEP2**

*Prof. Nick Mavromatos, Universitat de València, Burjassot, Spain, May 23 – 16, 2014*

*Lecture series*

No. of credits: 2

### **Dark Matter**

*Prof. Alejandro Ibarra, Universitat de València, Burjassot, Spain, September 23 – 27, 2013*

*Lecture series*

No. of credits: 2

### **International Cargèse School on Cosmic Accelerators**

*Institut d'Études Scientifiques de Cargèse, Cargèse, France, April 23 – May 8, 2013*

*Summer school*

### **Introduction to C++ Programming**

*Dr. Jacek Generowicz, Universitat de València, Burjassot, Spain, April 9 – 12, 2012*

*Workshop*

No. of credits: 6

### **Numerical Relativistic Astrophysics**

*Prof. Luciano Rezzolla, Universitat de València, Burjassot, Spain, March 27 – April 4, 2012*

*Lecture series*

No. of hours: 9

## Fortran for Scientific Computing

HLRS, University of Stuttgart, Stuttgart, Germany, Mar. 5 – 9, 2012

No. of hours: 33

Workshop

## Certifications and Credentials

---

### Gen AI Intensive

*Generative AI*

Kaggle

### Data Scientist Professional with Python

*Python Programming, Data Science, Data Communication, Machine Learning*

DataCamp

### Machine Learning Scientist with Python

*Machine Learning, NLP, Deep Learning, Image Processing, Big Data*

DataCamp

### Mathematical Foundations of Machine Learning

*Python Programming, Data Science, Statistics, Machine Learning*

Udemy

### Python for Statistical Analysis

*Python Programming, Data Science, Data Communication, Machine Learning*

Udemy

## Outreach

---

### Los más rápidos y los más furiosos (*The Fastest and the Most Furious*)

*Community of Undergraduate Physics Students, Juárez Autonomous University of Tabasco*  
Tabasco, Mexico

Online talk

September 4, 2020

### Una simulación de la física y la astrofísica (*A Simulation of Physics and Astrophysics*)

*Community of Undergraduate Physics Students, Juárez Autonomous University of Tabasco*  
Tabasco, Mexico

Online talk

August 14, 2020

### ANITA y la teoría de los universos paralelos (*ANITA and the Theory of Parallel Universes*)

*Científicos Mexicanos en el Extranjero, [mexiciencia.github.io/post/anita](https://mexiciencia.github.io/post/anita)*

Blog post

May 29, 2020

### ¿Qué es el modelo SIR? (*What is the SIR Model?*)

*Científicos Mexicanos en el Extranjero, [mexiciencia.github.io/post/modelo-sir](https://mexiciencia.github.io/post/modelo-sir)*

Blog post

May 25, 2020

### Evolución del brote epidémico de COVID-19 (*Evolution of the COVID-19 Epidemic Outbreak?*)

*Científicos Mexicanos en el Extranjero, [mexiciencia.github.io/post/covid19](https://mexiciencia.github.io/post/covid19)*

Blog post

April 5, 2020

Collaborator with the data analysis/modeling

### Annual Department of Physics and Astronomy Poster Event

*Department of Physics and Astronomy, Purdue University*  
West Lafayette, IN, USA

Posters (3) presentation

November 13, 2019

### Post-Doc Panel Q&A: What Happens When we Complete our PhDs?

*Department of Physics and Astronomy, Purdue University*  
West Lafayette, IN, USA

Panelist

April 10, 2019

### Annual Department of Physics and Astronomy Poster Event

*Department of Physics and Astronomy, Purdue University*  
West Lafayette, IN, USA

Poster presentation

November 14, 2018

### ¿Decía Einstein la verdad? (*Was Einstein Telling the Truth?*)

*Facultad de Ciencias, Universidad Autónoma del Estado de México*  
Toluca, Mexico

Talk

March 11, 2009

## Synergetic Activities

---

### X Scientific Meeting of the Spanish Astronomical Society

*Organizing contributor, 14–16 December, 2012*

Valencia, Spain

## Other activities

---

**Aug 2007– May 2009:** *Representative* of the Physics students community at the Governing Council of the Faculty of Sciences of the Universidad Autónoma del Estado de México.

## Languages

---

**Spanish:** Native proficiency

**Catalan:** Basic

**English:** Full professional proficiency

**Nahuatl:** Basic